Machine Learning Basics

Classification of AI, Supervised & Unsupervised Learning, and Beyond

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Part 1: Classification of AI

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Classification of AI

- **Artificial Narrow Intelligence (ANI)**: AI designed for a specific task (e.g., image recognition, playing chess). This is all of AI today.
- **Artificial General Intelligence (AGI)**: A hypothetical AI that can perform any intellectual task a human can. Does not exist yet.
- **Artificial Super Intelligence (ASI)**: A hypothetical AI that surpasses human intelligence in all areas. A topic of much debate.

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Approaches of AI (Non-Inclusive)

- **Symbolic / Logic-Based**: Uses rules and symbol manipulation (e.g., Expert Systems).
- **Statistical / Probabilistic**: Uses probability theory to make decisions (e.g., Bayesian Networks, Statistical Learning).
- **Computational (Connectionist)**: Inspired by the brain, uses neural networks (e.g., Deep Learning).
- **Integrative**: Combines multiple approaches.

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AI Research Topics

- Machine Learning
- Computer Vision
- Natural Language Processing (LLMs)
- Reinforcement Learning
- Knowledge Representation and Reasoning
- Planning and Scheduling
- Search Methodologies

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The AI / ML / DL Hierarchy

- **Artificial Intelligence (AI)**: The broadest field, encompassing all techniques that enable machines to mimic human intelligence.
- **Machine Learning (ML)**: A subset of AI. Systems that learn from data without being explicitly programmed.
- **Deep Learning (DL)**: A subset of ML. Uses multi-layered neural networks to learn complex patterns.

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Part 2: Introduction to Machine Learning

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What is "Learning" for a Machine?

- "Machine Learning is the science of getting computers to act without being explicitly programmed." — Arthur Samuel, 1959.
- Instead of writing rules, we provide **data** and let the machine **discover** the rules.

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The 4 Ingredients of Machine Learning

- **1. Data**: The raw material. The experience the machine learns from. (e.g., images, emails, stock prices).
- **2. Model**: The "brain" or the algorithm. A mathematical representation of the world. (e.g., a neural network, a decision tree).
- **3. Objective Function (Loss)**: The "teacher." A function that tells the model how well it's doing. The goal is to minimize this.
- **4. Learning Algorithm**: The method used to "train" the model to achieve the objective (e.g., Gradient Descent).

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Categorizing ML Techniques

- We categorize ML techniques based on the kind of **Data** and **Objective** they use.
- Main paradigms: Supervised, Unsupervised, Reinforcement Learning.

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Part 3: Supervised Learning

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What is Supervised Learning?

- **Definition**: The task of learning a function $f(X) \to y$ that maps an input X to an output y based on example input-output pairs.
- **Keywords**: Labeled data, training data, ground truth.

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The Supervised Learning Loop

- 1. **Input** X is fed into the **Model**.
- 2. Model produces an **Output** (prediction).
- 3. **Loss Function** compares output to the **Targeted Output** (ground truth).
- 4. **Learning Algorithm** adjusts the model based on the loss.
- 5. Repeat until the loss is minimized.

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Two Major Tasks of Supervised Learning

- **Classification**: Predicting a **discrete category**. Question: "What class does this belong to?" (e.g., spam vs. not spam, cat vs. dog).
- **Regression**: Predicting a **continuous value**. Question: "How much?" or "How many?" (e.g., predicting house prices, stock prices).

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Technique 1: Linear Regression

- **Goal**: Model the relationship between a dependent variable (y) and one or more independent variables (X) by fitting a straight line.
- The model learns the optimal slope and intercept to minimize the prediction error.

🚀 **Interactive Demo:** linear_regression_demo.html

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Technique 2: k-Nearest Neighbor (kNN)

- A simple, intuitive classification algorithm.
- **Step 1**: Find the k nearest data points (neighbors) to the query point.
- **Step 2**: Classify the query point by a **majority vote** of its neighbors.

🚀 **Interactive Demo:** knn_demo.html

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Part 4: Unsupervised Learning

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What is Unsupervised Learning?

- **Definition**: You only have input data (X) and no corresponding output labels. The goal is to find hidden patterns or intrinsic structures.
- **Keywords**: Unlabeled data, hidden structure, patterns.

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Classic Unsupervised Task: Clustering

- **Question it answers**: "What are the natural groups in my data?"
- **Examples**: Customer Segmentation, Image Segmentation, Genomics (grouping genes with similar expression patterns).

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Technique 3: K-Means Clustering

- A simple and popular clustering algorithm.
- **Step 1**: Randomly initialize K cluster centroids.
- **Step 2**: Assign each data point to the nearest centroid.
- **Step 3**: Update each centroid to be the mean of its assigned points.
- **Step 4**: Repeat steps 2-3 until convergence (centroids stop moving).

🚀 **Interactive Demo:** kmeans_demo.html

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Part 5: Learning with Less Supervision

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The Labeled Data Bottleneck

- Labeled data is the key ingredient for supervised learning, but it's expensive and time-consuming to create.
- The world is full of **unlabeled data** (all of Wikipedia, all of YouTube).
- How can we leverage this?

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Semi-Supervised Learning

- **Concept**: Use a small amount of labeled data + a large amount of unlabeled data.
- The model learns the overall structure from the unlabeled data to improve its decisions on the labeled data.
- **Analogy**: A student has an answer key for Chapter 1, but not for Chapters 2-20. They can still learn a lot from the text.

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Transfer Learning

- **Concept**: Take a model pre-trained on a massive dataset and adapt it for a more specific task.
- **Analogy**: A master chef doesn't need to re-learn how to hold a knife to learn a new recipe. They transfer their skills.
- **Supervised Fine-Tuning (SFT)**: The process of adapting a pre-trained model using your smaller, labeled dataset. Extremely common today.

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Self-Supervised Learning (The Key to LLMs)

- **Concept**: Create a supervised learning task from the unlabeled data itself.
- **Analogy**: Take a sentence from Wikipedia, hide a word, and ask the model to predict the missing word. The data is the sentence, the label is the hidden word!
- **Impact**: This is the core idea behind models like GPT and Gemini. They learn the structure of language by predicting the next word.

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Part 6: Reinforcement Learning

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What is Reinforcement Learning?

- It's not about labeled data; it's about learning to make a **sequence of decisions**.
- **Analogy**: Training a dog. You don't show it a flashcard. You say "sit," and when it sits, you give it a treat (reward). Over time, it learns a **policy** to maximize its treats.

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Key Components of RL

- **Agent**: The learner or decision-maker (the dog, a robot, a game-playing AI).
- **Environment**: The world the agent interacts with (a room, a maze, a game of Chess).
- **Action**: A move the agent can make.
- **Reward**: The feedback from the environment (positive or negative).
- **Policy**: The agent's strategy for choosing actions.

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RL Examples

- **Robotics**: Learning to walk or grasp objects.
- **Game Playing**: AlphaGo defeating the world's best Go player.
- **Resource Management**: Optimizing cooling systems in data centers (Google DeepMind).

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Summary

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Key Takeaways

- **Supervised Learning**: Labeled Data (Classification, Regression).
- **Unsupervised Learning**: Unlabeled Data (Clustering).
- **Reinforcement Learning**: No data labels, just an Environment (Agent, Rewards, Policy).
- **Semi/Self/Transfer Learning**: Powerful bridges that reduce the need for expensive labeled data.

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